

Question:

Calculate the quantity of heat required to (a) convert a liter of H₂O at 30°C to a liter of water at 60°C, and (b) heat a 1 kg block of aluminum from 30° to 60°C. Assume the specific heat of water and aluminum is, re-spectively, 1 cal/g°C and .215 cal/g°C.

Answer Choices:

- A. 28 Kcal, 6.3 Kcal
- B. 40 Kcal, 5 Kcal
- C. 30 Kcal, 8.58 Kcal
- D. 25 Kcal, 8 Kcal
- E. 29 Kcal, 7.77 Kcal
- F. 34 Kcal, 6.9 Kcal
- G. 30 Kcal, 6.45 Kcal
- H. 32 Kcal, 7 Kcal
- I. 35 Kcal, 7.5 Kcal
- J. 31 Kcal, 6.15 Kcal

Reasoning + Answer:

<think> To find the quantity of heat required for both (a) and (b), we need to calculate the temperature change for each and use the formula $Q = mc\Delta T$, where m is the mass , c is the specific heat capacity , and ΔT is the change in temperature .

For (a), a liter of water has a mass of approximately 1000 g (since 1 L of water has a density of approximately 1 g/mL), and the temperature change is from 30 °C to 60 °C , $\Delta T = 30 \text{ }^{\circ}\text{C}$. The specific heat of water is 1 cal/g °C .

$$Q_{\text{water}} = (1000 \text{ g})(1 \text{ cal/g }^{\circ}\text{C})(30 \text{ }^{\circ}\text{C}) = 30,000 \text{ cal} = 30 \text{ Kcal}$$

For (b), a 1 kg block of aluminum has a mass of 1000 g , the temperature change is also 30 °C , and the specific heat of aluminum is 0 . 215 cal/g °C .

$$Q_{\text{aluminum}} = (1000 \text{ g})(0 . 215 \text{ cal/g }^{\circ}\text{C})(30 \text{ }^{\circ}\text{C}) = 6435 \text{ cal} \approx 6 . 435 \text{ Kcal}$$

So , the answer for (a) is 30 Kcal , and for (b) is 6 . 45 Kcal .</think><answer> G </answer>

